Fiscal Year:	FY 2013	**Task Last Updated:**	FY 12/03/2012
PI Name:	Amundson, Sally Sc.D.		
Project Title:	HZE-induced mammary cancer development processes in murine and 'humanized' models, and their influence on radiation quality functions		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
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Zip Code:	10032	**Congressional District:**	15
Comments:			
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Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Ullrich, Robert (University of Texas Medical Branch)		
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Task Description:	The proposed research will use a model of breast cancer development in which mouse or human mammary stem cells are used to grow mammary glands in mouse mammary fat pads. When the stem cells used are irradiated prior to transplant, progression from very early stages of cancer (ductal dysplasias) can be monitored. We will use this model to compare the development of cancer from cells exposed to gamma-rays versus cells exposed to HZE particles that are found in galactic cosmic rays. The histological progression of pre-cancerous lesions will be monitored using markers of cell growth, senescence, apoptosis, proliferation and angiogenesis. This information will also be correlated with global gene expression profiling of non-persistent and persisting dysplasias to gain further insight into the molecular mechanisms driving progress towards cancer after exposure to the different types of radiation. Studying the cancers that are initiated by radiation in both mouse and human mammary cells will allow a direct comparison of cancer processes that may differ between species as well as between radiation types. This is important, as most assessments of human space radiation risk are based on experiments in rodents.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will provide insight into the cancer risks arising from exposure to space radiation. It asks questions about how cancer is started by radiation exposure, how cancer progresses, and how these processes might be the same and different after exposure to radiation encountered in space, and the more common gamma-rays encountered on earth. Since breast cancer is one of the most common cancers associated with radiation exposure, this is an important model system. Understanding the progression of breast cancer after exposure to space radiation will help NASA to make better risk estimates for their long-term exploration missions. A better understanding of the factors that control breast cancer progression in general can also have a large impact on Earth, by suggesting better intervention or treatment strategies for this very common disease.
Task Progress:	Annual Report (2/1/2012 – 1/31/2013). PI: Sally A. Amundson. Grant No: NNX09AE95G. Columbia University Medical Center 630 W. 168th St., VC11-230 New York, NY 10032 The goal of Specific Aim 1 is “to quantitatively compare early tumor initiating events induced by HZE vs. gamma rays in murine mammary epithelial stem cells using the well-established ductal dysplasia model as well as a 3D in vitro model.” We are using a murine intraductal transplantation model for the analysis of neoplastic progression in murine mammary epithelial cells derived from irradiated mice. In our model, intraductal injection of murine mammary epithelial cells allows quantification and temporal analysis of processes involved in early events in mammary carcinogenesis. Mammary epithelial tissue was harvested 24 hours after exposure of Balb/c mice to with SPE1972 Protons, 1 GeV/n 56Fe, 350 MeV/n 28Si, gamma rays, or from controls. Cells from the tissue were dissociated and grown in attachment-free and non-differentiating conditions to allow mammosphere formation. Mammospheres are enriched in progenitor cells capable of differentiating along luminal, myoepithelial, and alveolar lineages. Ten days after culturing in these conditions the primary mammospheres were collected, dissociated into individual cells, and cultured as secondary mammospheres. After another approximately 10 days these secondary mammospheres were collected and injected into recipient mice for the formation of dysplastic lesions. Samples of both the primary and secondary mammosphere cultures were retained for gene expression studies. While the first set of mice sacrificed (4 weeks) showed similar lesion frequencies to those that had been seen previously, at the second set of sacrifices (12 weeks) we found that all of the initial outgrowths had regressed when allowed to remain in situ. The reason for this discrepancy from previous results is not known, and the experiments are currently being repeated. The goal of Specific Aim 2 is “to assay both phenotypic progression and molecular profiles using the mammary ductal dysplasia and 3D stem cell models to determine whether HZE irradiation results in qualitatively different processes, compared with gamma-ray irradiation, that point to different mechanisms of carcinogenesis.” For these studies we are using the dysplasia assay to measure both total dysplasias as determined at around 4 weeks following intraductal injection of donor mammospheres and persistent dysplasias as quantified at around 12 weeks after injection as described under Specific Aim 1 above. RNA was harvested from lesions from mice sacrificed at 4 weeks. As noted above, for unknown reasons no lesions were identified in mice sacrificed at 12 weeks. Unfortunately, the RNA from the early lesions was of lower than expected quality, with RIN values between 4 and 6, indicative of degradation. In light of the low quality and missing late lesion samples, array hybridization was not undertaken and the experiment is being repeated. Due to the persistent technical difficulties encountered with the mouse lesions, we have extended our work to include studies of human cancer stem cells in support of our aims. Cancer stem cells (CSC) can be identified in several different tumors based on the presence or absence of specific cell surface markers. Breast CSCs have been identified by the surface marker expression of CD44+/CD24low. CD44 is expressed on the surface of cancer stem cells from a variety of tumor types and high CD44 expression is associated with increased risk of metastasis and poor overall patient prognosis. CD44 positive breast cancer stem cells have been shown to be highly refractory to radiation and chemotherapeutics and are enriched following radiation treatment. We exposed exponentially growing cultures of breast cancer cell lines (MDA-231, MDA-468, and MCF-7) to gamma radiation and analyzed CD44 expression 24 and 5 days post-exposure using flow cytometry. The standard CD44 form (CD44s) was measured and a change in CD44s intensity was observed in the cell lines 24 and 5 days after exposure to gamma radiation. In MDA-231 the pattern of CD44s intensity was decreased at a dose as low as 0.1 Gy and increased following 10 Gy radiation exposure as early as 24 hours and persisted after 5 days. MDA-231 and MDA-468 displayed similar CD44s patterns 5 days post-exposure. Interestingly, MCF7 showed decreased CD44s levels for each radiation dose at 24 hours but no change at 5 days post-exposure. The patterns of CD44s expression were stable for at least two weeks in progeny of irradiated cells at all doses studied. Taken together these data show that radiation alters CD44 signaling in breast cancer cells in a dose dependent manner. Mammalian tissues generally experience oxygen (O2) concentrations in the range of 2 to 8%, far lower than the standard atmospheric O2 environment (20%) in which most cells are cultured in the laboratory. When grown under atmospheric O2 many human cell lines have been shown to accumulate oxidative damage and undergo premature senescence. To test the effect of O2 culture conditions on the intensity of CD44s staining we cultured MDA-231, MDA-468, and MCF7 at 3% O2. The results at 24 hours post exposure revealed a different pattern of CD44s intensity compared to 20% O2 in each cell line. Radiation dose dependent differences were not observed in MDA-231 or MCF7.

	In MDA-468 increased CD44s intensity was observed after exposure to 10 Gy. At 5 days after radiation exposure the pattern of CD44s in MDA-231 was similar when grown under 3% or 20%. In MDA-468 cells grown at 3% O2 CD44s intensity decreased (relative to control) for both the 0.1 and 1.0 Gy samples. Although no radiation effect on CD44s intensity was observed in MCF7 cells grown at 20% O2, under 3% O2 they showed a CD44s pattern similar to MDA-231 and MDA-468. The data confirm our initial finding that low dose exposures seemed to decrease the "stemness" of cancer cell populations as defined by CD44 expression, while high doses increase the tendency to "stemness." It also indicates that O2 tension influences cancer stem cell physiology and radiation response, and should be considered when using in vivo models.
Bibliography Type:	Description: (Last Updated: 09/17/2020)
Abstracts for Journals and Proceedings	Meador JA, Yu Y, Ullrich RL, Amundson SA. "HZE-Induced Mammary Cancer Development Processes." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012